EXP 30 :

Aim

To write a Python program using Matplotlib to create a grouped bar chart showing the scores of men and women for five groups using multiple X-values on the same chart.

Algorithm

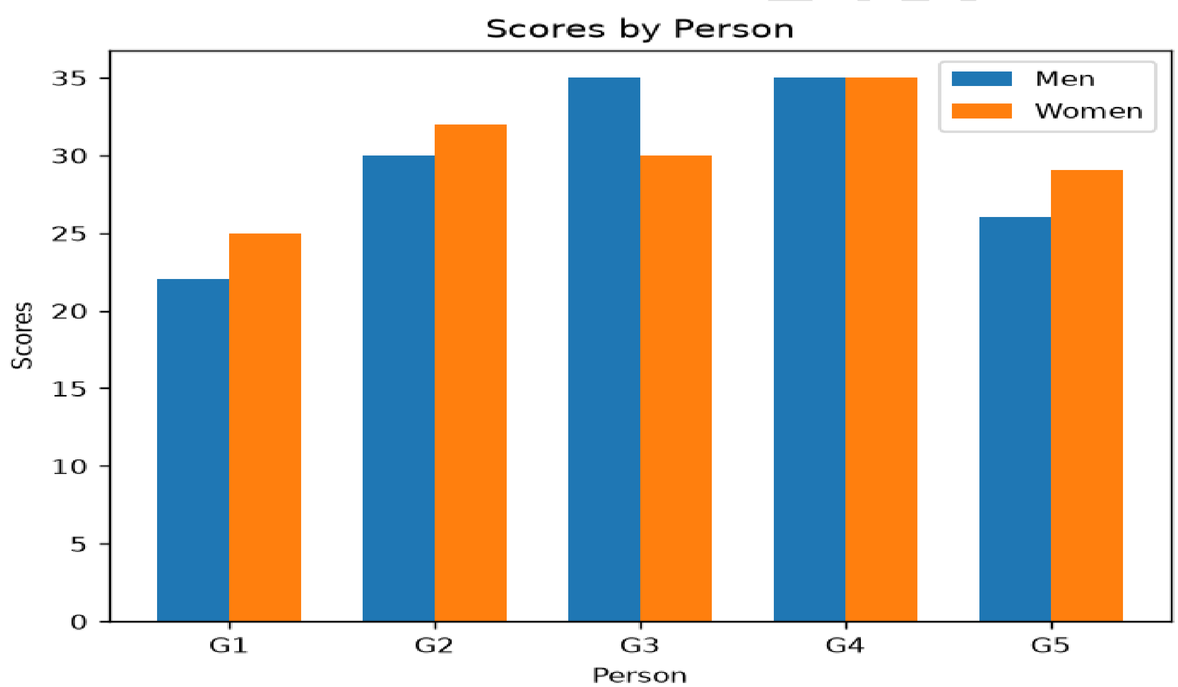
1. Import the matplotlib.pyplot library.
2. Define the scores of men and women.
3. Define the X-axis positions for the five groups.
4. Set the width of each bar.
5. Plot the bars for men and women side by side.
6. Add X-axis label, Y-axis label, title, and legend.
7. Display the bar chart.

Code

```
import matplotlib.pyplot as plt
# Sample data
men = [22, 30, 35, 35, 26]
women = [25, 32, 30, 35, 29]
# X-axis positions
x = [1, 2, 3, 4, 5]
width = 0.35
# Create bar chart
plt.bar([i - width/2 for i in x], men, width,
    label='Men')
plt.bar([i + width/2 for i in x], women, width,
    label='Women')
# Labels and title
plt.xlabel('Person')
```

```
plt.ylabel('Scores')

plt.title('Scores by Person')

# X-axis labels

plt.xticks(x, ['G1', 'G2', 'G3', 'G4', 'G5'])

# Legend

plt.legend()

# Display graph

plt.show()
```

Output:



Result

The program successfully generates a grouped bar chart comparing the scores of men and women for groups G1, G2, G3, G4, and G5 on the same chart.

EXP 31 :

Aim

To write a Python program using Matplotlib to create a stacked bar plot with error bars, where the women's bars are stacked on top of the men's bars.

Algorithm

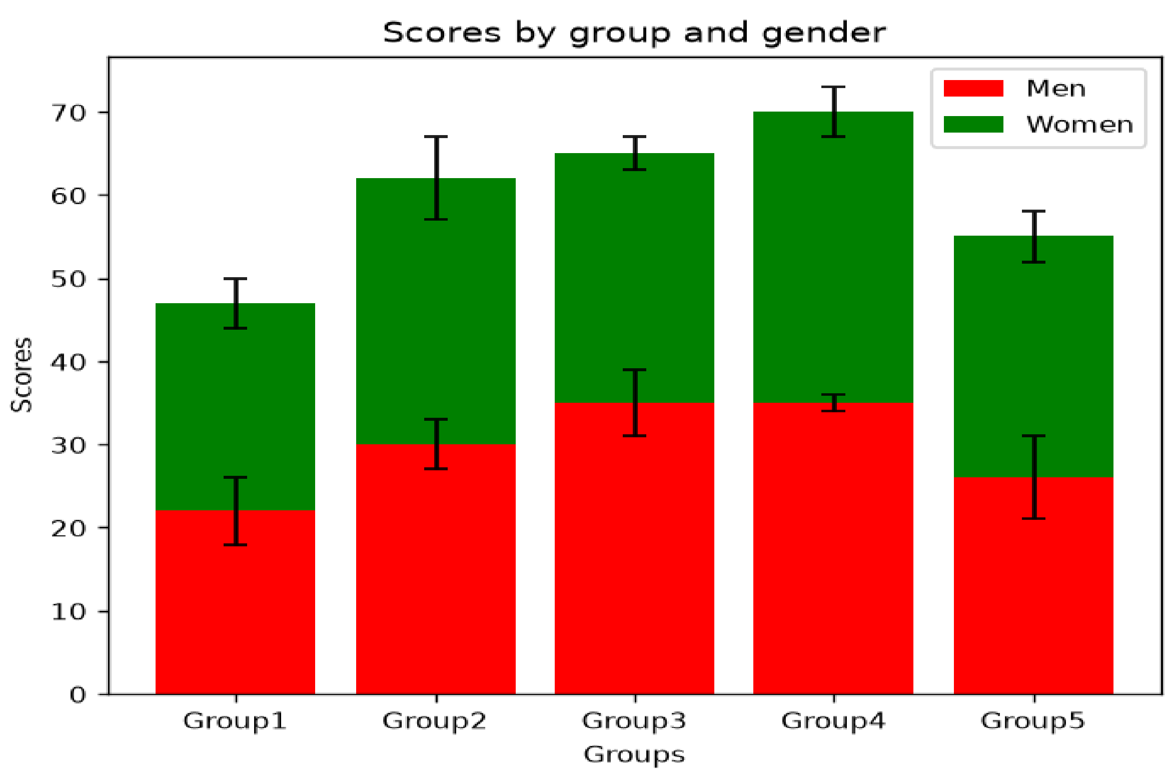
1. Import matplotlib.pyplot.
2. Define the mean scores for men and women.
3. Define the standard deviations for men and women.
4. Create positions for the five groups.
5. Plot the men's bars with their standard deviation as error bars.
6. Use bottom=men to stack the women's bars on top of the men's bars.
7. Add women's error bars.
8. Add labels, title, legend, and display the graph.

Code

```
import matplotlib.pyplot as plt

# Sample data

men = [22, 30, 35, 35, 26]

women = [25, 32, 30, 35, 29]

# Standard deviations

men_sd = [4, 3, 4, 1, 5]

women_sd = [3, 5, 2, 3, 3]

# X-axis positions

x = range(5)

# Plot men

plt.bar(x, men, yerr=men_sd, capsize=4,

    label='Men', color='red')
```

```
# Plot women on top of men
plt.bar(x, women, bottom=men, yerr=women_sd,
        capsize=4, label='Women', color='green')
# Labels and title
plt.xlabel('Groups')
plt.ylabel('Scores')
plt.title('Scores by group and gender')
# X-axis labels
plt.xticks(x, ['Group1', 'Group2', 'Group3', 'Group4', 'Group5'])
# Legend
plt.legend()
# Display graph
plt.show()
```

Output:



Result

The program successfully creates a stacked bar chart with error bars, where the women's scores are stacked on top of the men's scores for all five groups. The error bars represent the standard deviation of each group's scores.

EXP 32:

Aim

To write a Python program using Matplotlib to draw a scatter plot by generating random values for X and Y and plotting them against each other.

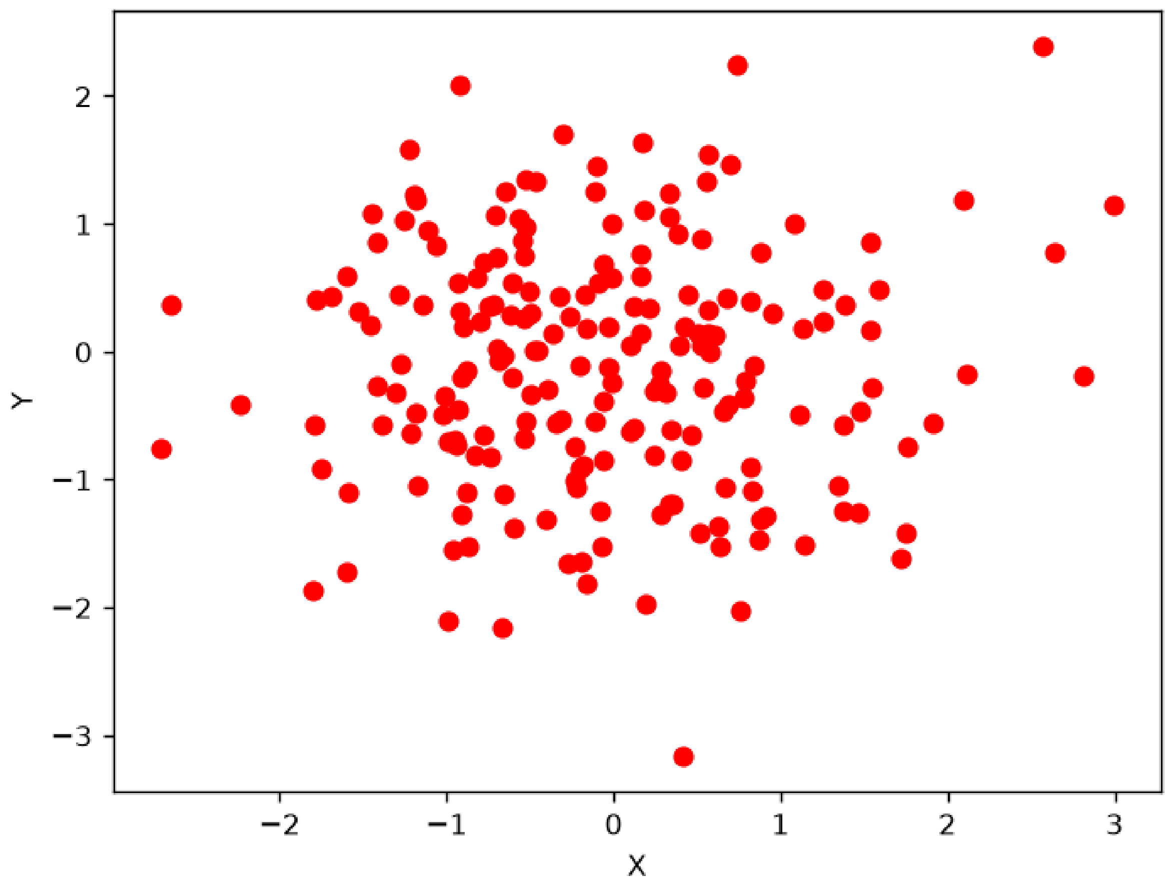
Algorithm

1. Import numpy and matplotlib.pyplot.
2. Generate random values for X and Y using a normal distribution.
3. Create a scatter plot using plt.scatter().
4. Label the X-axis and Y-axis.
5. Display the scatter graph.

Code

```
import numpy as np

import matplotlib.pyplot as plt

# Generate random values

x = np.random.randn(200)

y = np.random.randn(200)

# Create scatter plot

plt.scatter(x, y, color='red')

# Add labels

plt.xlabel('X')

plt.ylabel('Y')

# Display graph

plt.show()
```

Output:



Result

The program successfully generates a scatter graph containing randomly distributed values of X and Y, plotted against each other.